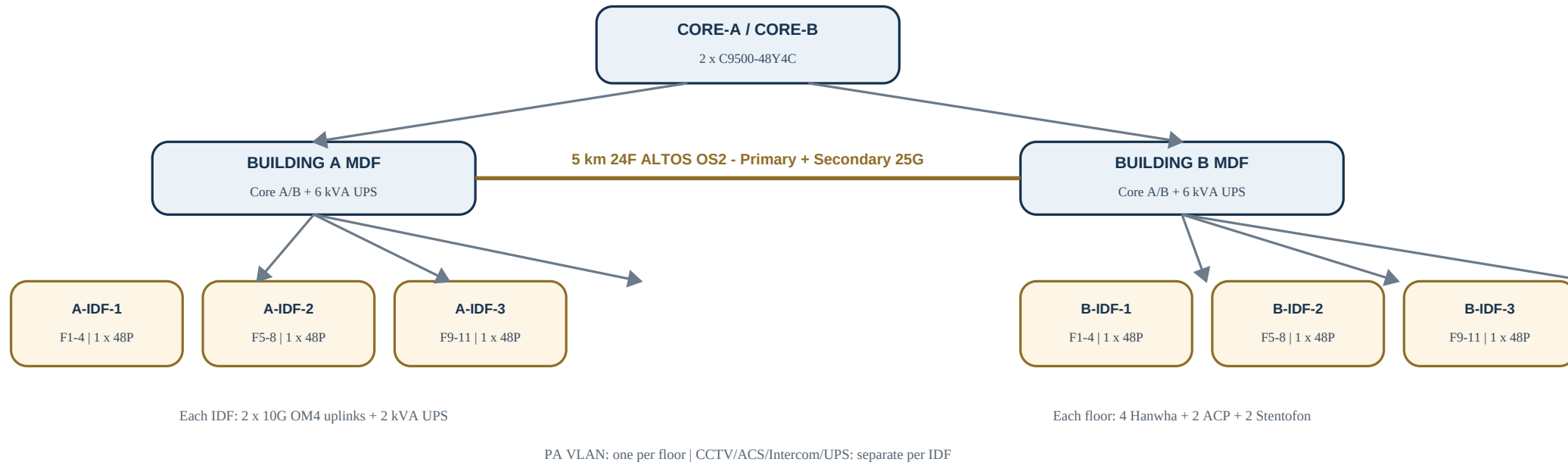


Two 11-Story Buildings - CCTV & ELV Network Design

Corning Fiber + Cat6A Structured Cabling + Cisco Switching + Hanwha CCTV + Stentofon VoIP Intercom



Portfolio scope: 2 buildings x 11 floors. One 48-port PoE switch per IDF (3 IDFs per building). Primary/secondary 10G IDF uplinks over Corning OM4 LSZH; primary/secondary 25G inter-building links over 5 km of 24F Corning ALTOS OS2.

Assumptions: 4 m floor-to-floor height; approximately 20 m Cat6A average endpoint run; "OMM" interpreted as OM4. PA endpoint quantity was not specified, so PA is VLAN-provisioned but not quantity-invented.

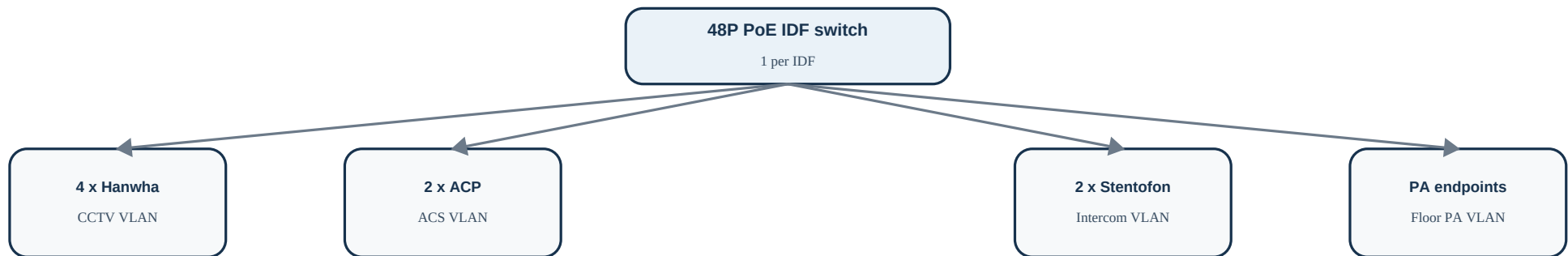
Design date: 27 September 2026

1. Design Requirements

Item	Building A	Building B	Total
Stories	11	11	22
IDFs	3	3	6
MDFs	1	1	2
Hanwha IP cameras	44	44	88
Stentofon / Zenitel VoIP intercoms	22	22	44
Access-control panels	22	22	44
48-port PoE switches - one per IDF	3	3	6
IDF UPS	3 x 2 kVA	3 x 2 kVA	6
MDF UPS	1 x 6 kVA	1 x 6 kVA	2
Primary + secondary 10G IDF uplinks	3 x 2	3 x 2	12 links
Primary + secondary 25G inter-building	2 links	2 links	2 links
Inter-building fiber	24F ALTOS, 5 km	-	5 km

IDF zoning: IDF-1 = Floors 1-4; IDF-2 = Floors 5-8; IDF-3 = Floors 9-11. Every IDF has a single 48-port PoE switch serving all floors in that IDF.

2. Floor / IDF Architecture



8 known networked endpoints per floor before PA: 4 CCTV + 2 ACP + 2 VoIP intercom

Each floor has 4 Hanwha cameras, 2 access-control panels and 2 Stentofon/Zenitel IP-SIP intercoms. These are home-run in Cat6A to the assigned IDF. The PA system receives its own VLAN per floor.

The IDF switch uplinks to MDF Core-A and Core-B using two independent 10G links over OM4. The access switch remains Layer 2 in this design; SVIs and inter-VLAN routing are located at the MDF core.

3. Fiber Backbone and Redundancy

Segment	Cable	Primary	Secondary	Optics	Termination
IDF -> MDF	12F OM4 LSZH	10G	10G	SFP-10G-SR	CCH-01U + LC UPC
Building A -> Building B	24F ALTOS OS2, 5 km	25G	25G	SFP-10/25G-LR-S	Configured Corning splice closure + core CCH

The 12F OM4 internal cable has ample fiber capacity for two duplex uplinks per IDF plus spare fibers. For a ground-level MDF and 4 m floor height, the estimated vertical distances to IDF-1/2/3 are about 16/32/44 m. With two links per IDF across two buildings, the theoretical vertical route total is about 368 m before installation slack and horizontal diversions; a 500 m procurement allowance is used in this portfolio.

Corning references a 12F OM4 LSZH loose-tube cable such as 012TUL-T4690D20. For the 5 km inter-building link, Corning ALTOS 24F OS2 options include 024EU4-T4100D20 and SMF-28 Ultra 024ZU4-T4122D20.

Fiber BOQ item	Qty / allowance	Reference P/N	Notes
Corning OM4 LSZH 12F internal cable	500 m	012TUL-T4690D20	Riser / indoor backbone; final length from coordinated drawings
Corning 1U CCH housing	8	CCH-01U	1 at each of 6 IDFs + 2 MDFs
CCH LC duplex UPC 6F OM4 panel	10	CCH-CP06-E4	1 per IDF, 2 per MDF as allowance
OM4 LC duplex pigtailed cassette	12 allowance	CCH-CS12-E4-P00QE	Final count depends on field-splice/panel configuration
10G SR MMF optics	24	SFP-10G-SR	12 IDF links x 2 ends
LC-LC OM4 duplex patch cords	48	Project-selected length	12 link ends/building x 2 buildings
Corning ALTOS 24F OS2 cable	5,000 m	024EU4-T4100D20 or 024ZU4-T4122D20	5 km route; two duplex pairs reserved for primary/secondary
25G LR-S SMF optics	4	SFP-10/25G-LR-S	2 inter-building links x 2 ends
Splice closures	2	Configured Corning FOSC / splice-closure family	One at each building entry / termination location

4. VLAN and IP Segmentation

The project uses different CCTV VLANs per IDF, separate ACS/intercom/UPS VLANs per IDF, and a separate PA VLAN per floor. The IP ranges below are an example private addressing plan for Packet Tracer and portfolio documentation.

Scope	Bldg	IDF/Floor	Function	VLAN	Example subnet
IDF-1	A	F1-4	CCTV	101	10.10.101.0/24
IDF-1	A	F1-4	ACS	201	10.11.201.0/24
IDF-1	A	F1-4	Intercom	301	10.12.301.0/24
IDF-1	A	F1-4	UPS	401	10.13.401.0/24
IDF-2	A	F5-8	CCTV	102	10.10.102.0/24
IDF-2	A	F5-8	ACS	202	10.11.202.0/24
IDF-2	A	F5-8	Intercom	302	10.12.302.0/24
IDF-2	A	F5-8	UPS	402	10.13.402.0/24
IDF-3	A	F9-11	CCTV	103	10.10.103.0/24
IDF-3	A	F9-11	ACS	203	10.11.203.0/24
IDF-3	A	F9-11	Intercom	303	10.12.303.0/24
IDF-3	A	F9-11	UPS	403	10.13.403.0/24
Floor 1	A	F1	PA	501	10.14.501.0/24
Floor 2	A	F2	PA	502	10.14.502.0/24
Floor 3	A	F3	PA	503	10.14.503.0/24
Floor 4	A	F4	PA	504	10.14.504.0/24
Floor 5	A	F5	PA	505	10.14.505.0/24
Floor 6	A	F6	PA	506	10.14.506.0/24
Floor 7	A	F7	PA	507	10.14.507.0/24
Floor 8	A	F8	PA	508	10.14.508.0/24
Floor 9	A	F9	PA	509	10.14.509.0/24
Floor 10	A	F10	PA	510	10.14.510.0/24
Floor 11	A	F11	PA	511	10.14.511.0/24
IDF-1	B	F1-4	CCTV	151	10.20.151.0/24
IDF-1	B	F1-4	ACS	251	10.21.251.0/24

Scope	Bldg	IDF/Floor	Function	VLAN	Example subnet
IDF-1	B	F1-4	Intercom	351	10.22.351.0/24
IDF-1	B	F1-4	UPS	451	10.23.451.0/24
IDF-2	B	F5-8	CCTV	152	10.20.152.0/24
IDF-2	B	F5-8	ACS	252	10.21.252.0/24
IDF-2	B	F5-8	Intercom	352	10.22.352.0/24
IDF-2	B	F5-8	UPS	452	10.23.452.0/24
IDF-3	B	F9-11	CCTV	153	10.20.153.0/24
IDF-3	B	F9-11	ACS	253	10.21.253.0/24
IDF-3	B	F9-11	Intercom	353	10.22.353.0/24
IDF-3	B	F9-11	UPS	453	10.23.453.0/24
Floor 1	B	F1	PA	551	10.24.551.0/24
Floor 2	B	F2	PA	552	10.24.552.0/24
Floor 3	B	F3	PA	553	10.24.553.0/24
Floor 4	B	F4	PA	554	10.24.554.0/24
Floor 5	B	F5	PA	555	10.24.555.0/24
Floor 6	B	F6	PA	556	10.24.556.0/24
Floor 7	B	F7	PA	557	10.24.557.0/24
Floor 8	B	F8	PA	558	10.24.558.0/24
Floor 9	B	F9	PA	559	10.24.559.0/24
Floor 10	B	F10	PA	560	10.24.560.0/24
Floor 11	B	F11	PA	561	10.24.561.0/24

Example gateway convention: the SVI for each VLAN uses .1 of that subnet (for example 10.10.101.1 for Building A IDF-1 CCTV). Access switches are trunks to the core; inter-VLAN routing is performed on Core-A/Core-B using SVIs.

5. Cat6A Cabling and Port Budget

Calculation	Per floor	22 floors	BOQ basis
Hanwha cameras	4 x 20 m = 80 m	1,760 m	Include 10% allowance
Stentofon VoIP intercom	2 x 20 m = 40 m	880 m	Include 10% allowance
ACP	2 x 20 m = 40 m	880 m	Include 10% allowance
Known total	160 m	3,520 m	3,872 m including 10%

Copper BOQ item	Quantity	Suggested reference	Notes
Cat6A permanent-link cable, LSZH, 23 AWG class EA	~3,872 m	Corning FutureCom / Everon Cat6A LSZH	Exact UAE/EMEA permanent-link P/N to be selected from approved regional catalogue
Cat6A endpoint ports / outlets	176	Corning Cat6A copper solution	8 known endpoints x 22 floors
Cat6A patch cords	176 minimum	Corning Everon CCAAGB family	1-2 m typical; final length by rack/device layout
Cable labels / management	1 lot	Project standard	At floor device and IDF rack ends

IDF port budget: a 4-floor IDF has 32 known networked endpoints (16 CCTV + 8 ACP + 8 intercom), leaving 16 copper ports on the 48-port switch before PA endpoints. A 3-floor IDF has 24 known endpoints, leaving 24 ports. Final PA port count and PoE draw must be added once the PA device quantity is defined.

6. Active Equipment and UPS

Equipment	Qty	Suggested model / part number	Role / reason
48-port PoE access switch	6	Cisco C9200L-48PXG-2Y-E	1 per IDF; 48 PoE+ ports; 2 x 25G uplinks usable with 10G optics
MDF/core Layer-3 switch	4	Cisco C9500-48Y4C	2 per MDF for Core-A/Core-B; 1/10/25G SFP28 ports
Hanwha IP camera	88	Hanwha QND-7082R	4MP IR dome representative model
Stentofon / Zenitel VoIP intercom	44	Zenitel TCIS-1, Item 1008111010	IP and SIP audio intercom
Access controller	44	Hikvision DS-K2624X	Four-door IP access controller
IDF UPS	6	APC SMT2200RMI2UC	2.2 kVA / 1.98 kW, 230V rackmount; used as 2 kVA class reference
MDF UPS	2	APC SRT6KXLI	6 kVA / 6 kW online UPS
10G SR optic	24	Cisco SFP-10G-SR	OM4 IDF primary/secondary links

Equipment	Qty	Suggested model / part number	Role / reason
25G LR-S optic	4	Cisco SFP-10/25G-LR-S	5 km OS2 primary/secondary inter-building links

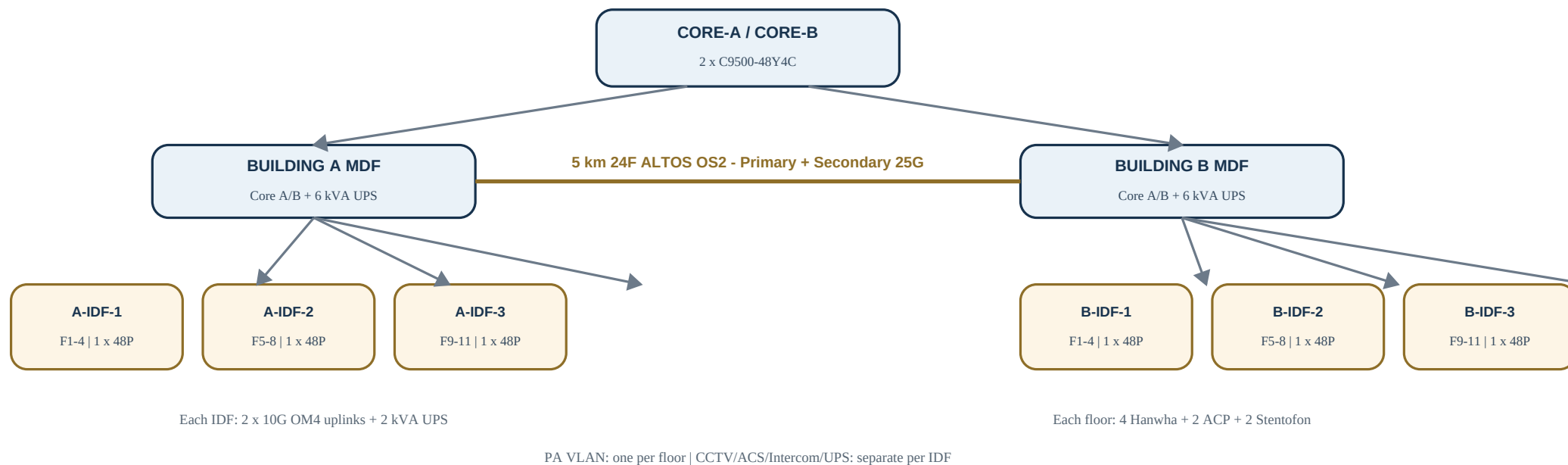
The selected C9200L-48PXG-2Y-E is a 48-port full PoE+ model with 2 x 1/10/25G fixed uplinks according to Cisco current documentation. The C9500-48Y4C provides 48 x 1/10/25G SFP28 interfaces, making it suitable for the core design.

7. PA VLANs and Inter-Building 25G Design

Item	Quantity	Reference	Treatment
Floor PA VLANs	22	Project VLAN plan	One dedicated PA VLAN on each floor
Example IP ceiling speaker	Design allowance only	Zenitel ZIPS-C, Item 1023320011	Use when the project specifies networked ceiling PA speakers; not counted in core endpoint total
PA controller / server	TBD	Zenitel platform dependent	Not counted because speaker count/zoning was not specified
Primary inter-building link	1 logical link	Core-A A <-> Core-A B	25G over one duplex pair of 24F OS2
Secondary inter-building link	1 logical link	Core-B A <-> Core-B B	25G over a second duplex pair of the same 24F OS2 cable

Redundancy note: Primary + secondary fiber pairs in the same 5 km cable provide fiber-pair/link redundancy. They do not provide path diversity. True route redundancy would require separate ducts/pathways and preferably separate cables.

8. End-to-End Flowchart



Signal flow: Floor devices -> Cat6A (~20 m) -> IDF 48P PoE switch -> dual 10G OM4 uplinks -> MDF Core-A/Core-B -> SVI/inter-VLAN routing/security policy -> CCTV/ACS/Intercom/PA servers. Building A Core-A/Core-B -> dual 25G OS2 -> Building B Core-A/Core-B.

9. Cisco Packet Tracer / Commissioning Test Plan

Test	Expected result
VLAN creation	Different CCTV VLAN for each IDF; separate ACS, intercom and UPS VLANs per IDF; separate PA VLAN on each floor
Trunk validation	show interfaces trunk shows correct allowed VLANs on every IDF uplink
SVI status	show ip interface brief shows all required SVI interfaces up/up
Routing table	show ip route shows connected VLAN networks and any configured dynamic/static routes
Inter-VLAN ping	Approved test endpoints can reach permitted services across VLANs
Primary IDF uplink failure	Traffic continues over secondary 10G uplink
Inter-building primary failure	Building-to-building traffic continues over secondary 25G link
PoE	Selected cameras/intercoms receive PoE and register
Fiber acceptance test	OLTS / OTDR verifies insertion loss, polarity and splice quality

10. Master BOQ Summary

Category	Description	Total Qty	Primary reference / model
Switching	48-port PoE IDF switch	6	Cisco C9200L-48PXG-2Y-E
Switching	Layer-3 core switch	4	Cisco C9500-48Y4C
CCTV	Hanwha 4MP IP dome cameras	88	QND-7082R
Intercom	Stentofon / Zenitel IP-SIP intercoms	44	TCIS-1 / 1008111010
ACS	4-door access controllers	44	Hikvision DS-K2624X
UPS	IDF UPS - 2 kVA class	6	APC SMT2200RMI2UC
UPS	MDF UPS - 6 kVA	2	APC SRT6KXLI
Fiber	OM4 LSZH 12F internal cable	500 m allowance	012TUL-T4690D20
Fiber	1U CCH housings	8	CCH-01U
Fiber	LC duplex UPC OM4 6F CCH panels	10	CCH-CP06-E4
Fiber	OM4 LC duplex pigtailed cassettes	12 allowance	CCH-CS12-E4-P00QE
Fiber	10G SR optics	24	Cisco SFP-10G-SR
Fiber	OM4 LC-LC duplex patch cords	48	Project-selected length
Fiber	24F ALTOS OS2 inter-building cable	5,000 m	024EU4-T4100D20 / 024ZU4-T4122D20
Fiber	25G LR-S optics	4	Cisco SFP-10/25G-LR-S
Fiber	Splice closures	2	Configured Corning FOSC / closure family
Copper	Cat6A horizontal cable	~3,872 m	Corning FutureCom / Everon Cat6A LSZH; regional P/N confirm
Copper	Known endpoint runs	176	4 camera + 2 ACP + 2 intercom per floor
PA	Dedicated PA VLANs	22	One per floor

11. Engineering Notes and Manufacturer References

Product references were checked against manufacturer documentation available at the design date. Regional/UAE commercial P/Ns can vary by cable jacket, fire rating, packaging length, licensing, power supply and market. Confirm final procurement codes with Corning/Cisco/Hanwha/Zenitel/Hikvision/Schneider before issuing a purchase order.

Corning CCH-01U is a 1U housing that accepts up to two CCH panels/cassettes. Corning lists LC duplex UPC panels including CCH-CP06-E4 for 50 um OM3/4 and LC UPC single-mode options such as CCH-CP24-A9.

Cisco documents SFP-10G-SR for OM3/OM4 MMF at up to 300/400 m and SFP-10/25G-LR-S for SMF up to 10 km. The 5 km inter-building link therefore uses LR-S optics subject to the project optical budget.

Zenitel lists TCIS-1 as an IP and SIP intercom (Item 1008111010) and ZIPS-C as an IP ceiling speaker (Item 1023320011). Hanwha lists QND-7082R as a 4MP IR dome camera. Hikvision lists DS-K2624X as a four-door IP access controller. Schneider lists SMT2200RMI2UC as a 2200 VA rackmount UPS and SRT6KXLI as a 6 kVA online UPS.

Manufacturer	Reference
Corning	ALTOS 24F OS2: 024EU4-T4100D20 / 024ZU4-T4122D20; OM4 LSZH 12F: 012TUL-T4690D20; CCH-01U; CCH-CP06-E4
Cisco	C9200L-48PXG-2Y-E; C9500-48Y4C; SFP-10G-SR; SFP-10/25G-LR-S
Hanwha Vision	QND-7082R
Zenitel / Stentofon	TCIS-1 Item 1008111010; ZIPS-C Item 1023320011
Hikvision	DS-K2624X
APC / Schneider Electric	SMT2200RMI2UC; SRT6KXLI

GitHub package: upload this PDF, the Cisco Packet Tracer .pkt file, a final topology PNG, a Cisco configuration text file and a README.md explaining the assumptions, VLAN plan, redundancy and test results.